YEAR 10 PHYSICS, AQA GCSE, LESSON 11 OF 12

ELECTRIC CIRCUITS

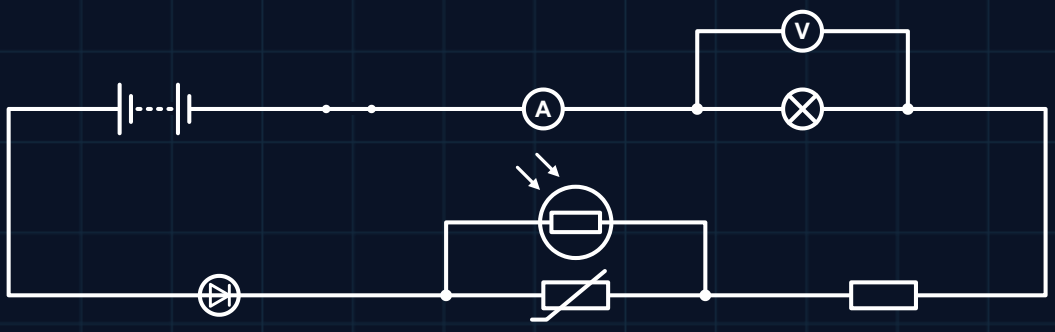
LESSON 11 POWER AND ENERGY TRANSFER

Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?

IN THIS LESSON YOU WILL

- describe power as the energy transferred per second
- use $E = P\,t$, $P = V\,I$ and $P = I^2\,R$ with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 11** of the Electric Circuits unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
The explanation, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own.

LAB
Record measurements from a virtual lab or a required practical.

EXAM CORNER
An AQA-style question with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the “Cover the page” questions in Revise it. (3) Check with the quick answers at the back.

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01	Complete circuits and symbols	02	Current and charge
03	Potential difference	04	Resistance and V = I R
05	Resistance of a wire RP	06	Series and parallel circuits
07	Resistors in series and parallel RP	08	I–V graphs
09	Investigating I–V characteristics RP	10	Thermistors, LDRs and sensors
11	**Power and energy transfer** Power, E = P t, P = V I, P = I² R, energy transfers in appliances	12	Mains, safety and the National Grid

Lesson

11 POWER AND ENERGY TRANSFER

AQA Combined 6.2.4.1–6.2.4.2; Physics 4.2.4.1–4.2.4.2

BIG QUESTION **Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?**

BY THE END YOU CAN

- describe power as the energy transferred per second
- use E = P t, P = V I and P = I² R with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

power **watt (W)** **kilowatt (kW)** **energy transferred** **power rating**

RETRIEVE

DO NOW

From earlier lessons. No notes!

1

5.0 C passes through a 12 V lamp. How much energy is transferred?

E = Q V = 60 J

2

2.0 A flows for 1 minute. How much charge flows?

Q = 2.0 × 60 = 120 C

3

What is the resistance of a heater with 230 V across it and 5.0 A through it?

R = 230 ÷ 5.0 = 46 Ω

EXPLAIN

HOW FAST IS ENERGY TRANSFERRED?

Power is the **energy transferred per second**. It is measured in **watts (W)**: 1 W = 1 J per second. A 2000 W (2 kW) kettle transfers 2000 J of energy every second.

APPLIANCE	TYPICAL POWER	ENERGY TRANSFERRED EACH SECOND
LED lamp	10 W	10 J
Laptop	60 W	60 J
Hairdryer	1.2 kW = 1200 W	1200 J
Kettle	2.2 kW = 2200 W	2200 J
Electric shower	9 kW = 9000 W	9000 J

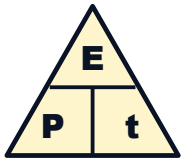
Every appliance transfers energy electrically from the mains to other energy stores: a kettle to the **thermal store** of the water; a hairdryer's fan to **kinetic** energy and its heater to thermal; a phone charger to the **chemical store** of the battery.

E = P t

energy transferred = power × time

E in joules (J), P in watts (W), t in seconds (s)

Convert kW → W (× 1000) and minutes or hours → seconds first.



WORKED EXAMPLE

A 2.0 kW kettle is on for 3.0 minutes. Calculate the energy transferred.

KNOW	P = 2000 W, t = 180 s
EQUATION	E = P t
SUBSTITUTE	E = 2000 × 180
ANSWER	**E = 360 000 J**

YOUR TURN

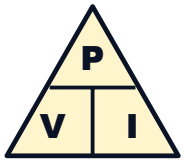
A 1.2 kW hairdryer is used for 10 minutes. Calculate the energy transferred.

KNOW	**P = 1200 W, t = 600 s**
EQUATION	**E = P t**
SUBSTITUTE	**E = 1200 × 600**
ANSWER	**E = 720 000 J**

P = V I

power = p.d. × current

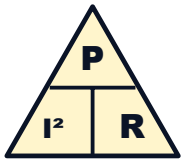
P in watts (W), V in volts (V), I in amperes (A)



P = I² R

power = current² × resistance

P in watts (W), I in amperes (A), R in ohms (Ω)



WORKED EXAMPLE	YOUR TURN
A heater on the 230 V mains draws a current of 4.0 A. Calculate its power.	**A current of 3.0 A flows through a 20 Ω resistor. Calculate the power.**
EQUATION P = V I	EQUATION **P = I² R**
SUBSTITUTE P = 230 × 4.0	SUBSTITUTE **P = 3.0² × 20 = 9.0 × 20**
ANSWER **P = 920 W**	ANSWER **P = 180 W**

! In P = I² R, **square the current first**. Doubling the current makes the heating **four times** bigger: that is why cables carrying big currents get hot.

IF YOU KNOW…	USE
p.d. and current	P = V I
current and resistance	P = I² R
power and time	E = P t
charge and p.d.	E = Q V
current and time	Q = I t

PRACTISE

TRY IT

1 Answer the big question. Show both calculations.

Kettle: E = 3000 × 180 = 540 000 J. LED: t = 5 × 3600 = 18 000 s, E = 10 × 18 000 = 180 000 J. The kettle transfers 3 times as much energy.

2 **SUPPORTED** A 12 V car headlamp draws a current of 5.0 A. Calculate its power.

EQUATION	**P = V I**	
SUBSTITUTE	**P = 12 × 5.0**	
ANSWER	**60 W**	unit:

3 **SUPPORTED** A 9.0 kW electric shower runs for 8.0 minutes. Calculate the energy transferred in joules.

CONVERT	**P = 9000 W, t = 480 s**	
EQUATION	**E = P t**	
SUBSTITUTE	**E = 9000 × 480**	
ANSWER	**4 320 000 J**	unit:

4 **ON YOUR OWN** A current of 2.0 A flows through a 10 Ω resistor. Then the current is increased to 4.0 A. Calculate the power each time. What happens to the power when the current doubles?

$P = 2.0^2 \times 10 = 40$ W; $P = 4.0^2 \times 10 = 160$ W. The power is 4 times bigger.

5 **ON YOUR OWN** A device transfers 240 J of energy when 40 C of charge flows through it. Calculate the p.d. across it.

V = E ÷ Q = 240 ÷ 40 = 6.0 V

6 **CHALLENGE** A lamp is rated 60 W, 230 V. Calculate the current through it and its resistance.

I = P ÷ V = 60 ÷ 230 = 0.26 A. R = V ÷ I = 230 ÷ 0.26 ≈ 880 Ω.

Get the marks: Write the power in watts before you substitute: 2.2 kW = 2200 W. Give answers to a sensible number of significant figures.

7 A 2.2 kW kettle runs on the 230 V mains. (a) Calculate the current through the kettle. (b) Calculate the energy transferred in 150 s. (c) A heater draws twice this current through the same resistance. How does its heating power compare? **[6 marks]**

(a) P = 2200 W (1); I = P ÷ V = 2200 ÷ 230 (1) = 9.6 A (1). (b) E = P t = 2200 × 150 (1) = 330 000 J (1). (c) P = I^2 R, so it is 4 times the power (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Power is energy transferred per second. 1 W = 1 J/s.
- **E = P t** (P in W, t in s).
- **P = V I** and **P = I² R**.
- Doubling the current through a resistor makes the heating 4 times bigger.
- Appliances transfer energy electrically to thermal, kinetic or chemical stores.

COVER THE PAGE AND ANSWER

1 What is power?

2 Convert 3.5 kW into watts.

3 Write two equations for electrical power.

4 Write the equation linking energy, power and time.

I CAN…

- describe power as energy transferred per second
- use E = P t with unit conversions
- use P = V I and P = I² R
- choose the right equation from the data given

REFERENCE

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: AQA gives you an equation sheet in the exam (it does not include $R_{total} = R_1 + R_2$), but knowing the equations by heart makes every question faster.

$Q = I\,t$

Lesson 2

charge flow = current × time

C, A, s

Rearranged: I = Q ÷ t and t = Q ÷ I

$E = Q\,V$

Lesson 3

energy transferred = charge × p.d.

J, C, V

Rearranged: V = E ÷ Q and Q = E ÷ V

$V = I\,R$

Lesson 4

p.d. = current × resistance

V, A, Ω

Rearranged: I = V ÷ R and R = V ÷ I

$R_{total} = R_1 + R_2$

Lesson 6

total resistance in series

Ω

Rearranged: (series only)

$P = V\,I$

Lesson 11

power = p.d. × current

W, V, A

Rearranged: V = P ÷ I and I = P ÷ V

$P = I^2\,R$

Lesson 11

power = current² × resistance

W, A, Ω

Rearranged: R = P ÷ I² and I = √(P ÷ R)

$E = P\,t$

Lesson 11

energy transferred = power × time

J, W, s

Rearranged: P = E ÷ t and t = E ÷ P

FOUR STEPS FOR EVERY CALCULATION

1
Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2
Write the equation you will use, in symbols.

3
Substitute the numbers, then rearrange if needed.

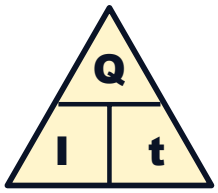
4
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes → seconds	× 60	3.0 min = 180 s
hours → seconds	× 3600	2.0 h = 7200 s
milliamps (mA) → amps (A)	÷ 1000	250 mA = 0.25 A
kilowatts (kW) → watts (W); kilohms (kΩ) → ohms (Ω)	× 1000	2.2 kW = 2200 W
kilovolts (kV) → volts (V)	× 1000	400 kV = 400 000 V

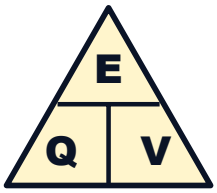
EQUATION TRIANGLES: HELP WITH REARRANGING

1. Cover the quantity you want to find.
2. The two left **side by side**? **Multiply** them.
3. One **above** the other? **Divide** the top by the bottom.

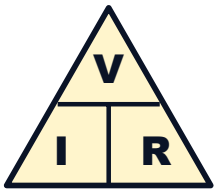
Example: to find I from Q = I t, cover I. Q is above t, so I = Q ÷ t. Triangles only help with rearranging: you still need to learn each equation. For P = I² R, the triangle gives I²; take the square root to find I.



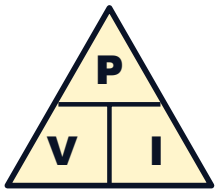
Q = I t
Q = I × t
I = Q ÷ t
t = Q ÷ I



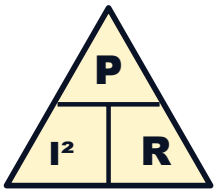
E = Q V
E = Q × V
Q = E ÷ V
V = E ÷ Q



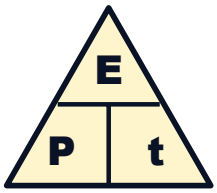
V = I R
V = I × R
I = V ÷ R
R = V ÷ I



P = V I
P = V × I
V = P ÷ I
I = P ÷ V



P = I² R
P = I² × R
R = P ÷ I²
I² = P ÷ R, so I = √(P ÷ R)



E = P t
E = P × t
P = E ÷ t
t = E ÷ P

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

Power (P): Energy transferred per second. Unit: watt (W).

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

DO NOW

1. E = Q V = 60 J
2. Q = 2.0 × 60 = 120 C
3. R = 230 ÷ 5.0 = 46 Ω

COVER THE PAGE

1. Energy transferred per second.
2. 3500 W
3. P = V I and P = I^2 R
4. E = P t

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